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Digital Selective Reversible Phase Control of Monolithically Integrated Heterogeneous Piezoelectric Polymer for Frequency Dependent Unimorph

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Thanks to spontaneous polarization in molecular structure, piezoelectric polymer, poly(vinylidene fluoride) (PVDF) holds great potential for diverse applications such as organic memory and electromechanical devices. However, the transformation of PVDF into a highly polarized β-phase has still relied on conventional processes such as repeated mechanical strain, high-temperature heat treatment, and high-voltage electric poling, which are time-consuming and can potentially cause undesired damages. Here, an ultrafast and reversible digital patterning process to transform the polymorphic phase of the PVDF has been developed using the interaction of laser with molecular structure. Plasmonic gold nanoparticles realize the interaction between PVDF and laser by increasing the absorption of the laser and amplifying its characteristics. The parameters of the laser process for phase conversion are designed under the theoretical background based on molecular dynamics (MD) simulation, and through this, the process is able to freely convert phases by simple parameter modifications. The selective laser process enables a monolithically integrated heterogeneous phase of PVDF which is not allowed in conventional single-phase producing processes. Moreover, a practical soft robot that can control its direction has been developed by utilizing the difference in mechanical responses of each phase to the electric field in a monolithically integrated single functional layer.

1. Introduction

Piezoelectric polymers show great potential for various electromechanical applications in soft platforms such as soft sensors, actuators, and energy harvesters owing to their spontaneous polarization in the chemical structure.[1–5] Poly(vinylidene fluoride) (PVDF) is the most popular piezoelectric polymer thanks to high electromechanical conversion efficiency, fast response, and high mechanical stability.[1–5] There are three well-defined semi-crystalline polymorphs of the PVDF which are α-phase with nonpolarity and β- and γ-phase with polarity.[6,7] Since the conversion of electric charge and mechanical energy is greatly determined by the polarization of the molecular structure,[8] techniques to create a largely polarized structure (i.e., β-phase) have been intensively studied.[8–20]

Most of the previous studies applied mechanical deformation (e.g., stretching or pressing) with heat treatment to change

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the molecular structure of PVDF.[11–15] High electric field poling techniques have also been studied a lot to change the structure of the PVDF by aligning the electric dipoles in the microstructure.[13,20] Although these processes have been well-established through the decades, they impose two major problems bringing a substantial inefficiency to the relevant research areas. First, the complexity of the multi-step process increases processing time and cost (e.g., biaxial stretching with thermal annealing, at least 40 cycles of high field poling)[13] but also lacks reproducibility since the entire process requires skilled manual labor (e.g., repeating several cycles that consists of hot pressing, cold water quenching, folding, cold pressing, annealing, hot pressing, and cold water quenching).[15] Second, several cycles of mechanical deformation and high-temperature annealing are inapplicable to a thin layer of PVDF, whereas most electronic devices require a thin and precise layer. Such processes inevitably damage and cause defects[12] to the PVDF and are also incompatible for electronics fabrication, where other comparts (e.g., electrode, substrate) have to be assembled in between. Therefore, a noble process of converting PVDF into β-phase is required that minimizes damage to film and is compatible with other functional layer forming processes for reliable and robust electronics applications.

Furthermore, despite efforts to improve electromechanical response through the high ratio of β-phase, it cannot lead to the development of electromechanical devices with practical and diverse functions since single-phase shows monotonous reactivity to external stimuli. For instance, the soft actuator remains at the level of realizing a simple vibrating cantilever or straight-moving robot without controllability.[5,21] Overcoming the conventional single-phase producing processes, researchers tried to pattern the ferroelectric property in a single PVDF layer.[22–27] Spatially controlled crystalline phases with different polarization (e.g., α-, β- and γ-phase) could produce strengthened or programmed reactivity to external stimuli[22,26] that can be applied to a high density of arrayed electronic devices such as sensors or memory devices. Furthermore, it also can achieve evolved and programmed responses to external stimuli through the difference in reactivity of each phase in a single PVDF which is not allowed in conventional single-phase producing processes.[25] However, previous studies also have several critical limitations. First, a microprinting technique achieved spatially controlled phases by locally applying pressure through the micropatterned mold.[22,23] But, such techniques require complex processes to fabricate the mold and time-consuming replicating procedures. Second, laser processes achieved selective phase patterning of crystalline phase, but it rather destroyed the polarization of molecular structure (i.e., polar β- to nonpolar α-phase transition) through the high energy of the laser.[24,25] Extreme laser condition to randomize the polarization was so slow that a long processing time was required and film destruction was observed. Recently, studies using plasmonic nanoparticles to increase laser absorption and changes in the phases of PVDF have been reported,[26,27] however, the level of phase transition was poor. Moreover, no studies on the laser-driven phase transition mechanism on a molecular scale were investigated. Demonstration of practical application through local phase patterning of piezoelectric PVDF is also expected but, has not been reported yet.

Here, we developed a novel, ultrafast, and programmable phase control process of the PVDF that utilizes the laser-induced phase transition (LIPT) phenomenon. We took advantage of the unique characteristics of the continuous wave (CW) laser, which can supply a strongly polarized electric field as well as photothermal energy in an ultrafast time scale to change the molecular structure of PVDF. Plasmonic nanomaterial was utilized to enhance the absorption of the laser through the localized surface plasmon resonance phenomenon. Such nanomaterials have played an important role in the engineering field that tried a local field enhancement of the light source (e.g., Raman scattering, photoluminescence).[28,29] Our study used gold nanoparticles (AuNPs) to amplify the electric field and thermal effects induced by the laser. Despite ultrafast laser scan speed (optimum scan speed: 1600 mm s^{-1}), thanks to nanoparticles, enough energy could be delivered to change the molecular structure of PVDF without surface damage. By controlling the laser parameters (e.g., power and scan speed), the LIPT not only produced polar β-phase in a fast, facile, and reliable manner but also reversibly converted the crystalline phases of the PVDF (reversible transition of α-phase $\leftrightarrow$ β-phase). Moreover, the phase transition mechanism was interpreted through molecular dynamics (MD) simulations. We finally demonstrated a next-generation electromechanical soft robot with a practical function capable of on-demand direction control.

2. Results and Discussion

2.1. Laser-induced Phase Transition of Ferroelectric Polymer, PVDF

The crystalline phases of the PVDF are determined by the combination of trans (T) and gauche (G: gauche, and G′: gauche-) conformers and have specific chain conformations[6,7] (TGTG′ for α-, TTTT for β-, and TTTGTTTG′ for γ-phase) (**Figure 1**a and Figure S1, Supporting Information). The difference in the polarization for each combination produces diverse electromechanical responses.[8] Outstripping the previous studies[8–20] that simply increased the single phase of the PVDF, the LIPT was able to control the most well-defined polymorphic crystalline phases (i.e., α-, β-, and γ-phase) of PVDF with high precision. AuNP inks (0.1 wt% in ethanol, PVP-stabilized, diameter = 40–60 nm) were spin-coated at 1,000 rpm to form a thin layer of plasmonic AuNPs on the prepared γ-phase PVDF to initiate the LIPT (Figure 1a). It showed strong absorption in the 500–600 nm wavelength region (Figure S2, Supporting Information). Such strong absorption was able to bring a synergetic effect with a 532 nm wavelength of CW laser. The transmittance of the PVDF with and without AuNPs showed an obvious difference in light absorption (Figure S3, Supporting Information). No specific change in the molecular structure of PVDF was observed by merely coating AuNPs (Figure S4, Supporting Information).

The highly polarized β-phase was produced through ultrafast laser scanning (Denoted by the LIPTβ, optimum laser condition: 800 mW, 1600 mm s^{-1}) on the prepared γ-phase with aid of AuNPs. The nonpolarized α-phase was obtained by simply changing the scan speed (Denoted by the LIPTα, optimum

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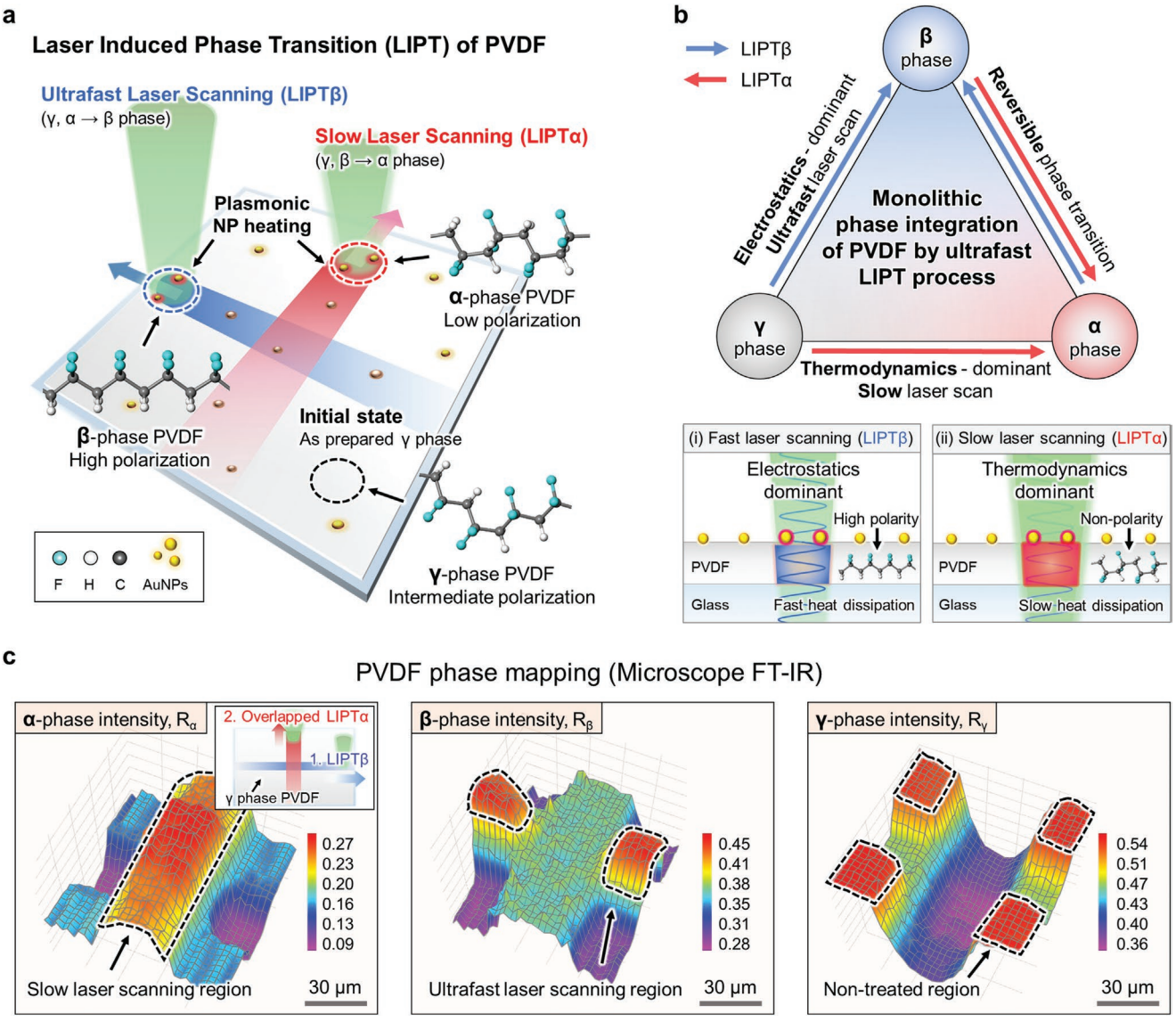


Figure 1. Laser-induced phase transition of ferroelectric polymer. a) Schematic illustration of the LIPT, laser-induced β-phase formation is depicted as blue color (LIPTβ) and laser-induced α-phase formation as red color (LIPTα). b) The schematics of process flow and significant advancement of LIPT (top), which enabled ultrafast and reversible monolithic phase integration, the explanation of laser effects on PVDF microstructure (bottom). c) FT-IR mapping images of monolithically integrated three polymorphic phases (α-, β-, and γ-phase), inset schematics shows sample preparation procedure. The mapping area in (c) was 100 μm × 100 μm.

laser condition: 800 mW, 1 mm s^{-1}). Little morphology change was observed after the ultrafast laser scan, whereas definite changes in surface morphology were found after slow scan speed. However, it did not induce significant damage to PVDF films (Figures S5 and S6, Supporting Information). AuNP is faithful to the role of amplifying the laser, but the particle itself does not affect the phase change during laser scanning since it has no polarity on the particle surface. In the case of LIPTβ, no changes in the size and contents of AuNPs were noticed before/after the laser process (Size ≈50–60 nm, distribution of 300–400 nm intervals). In the case of LIPTα, AuNPs were ablated due to an excessive supply of photothermal energy by low scan rates. More specifically, in the case of the LIPTβ, we hypothesized that PVDF was subjected to mild heat treatment with a strong AC electric field. As the laser beam moves, an ultrafast laser scan makes an electrostatics-dominant environment where moderate electric field and thermal energy were supplied simultaneously. The proper photothermal energy makes the backbone of the PVDF flexible enough, then the AC electric field gives a chance for PVDF molecules to be rearranged (Figure 1bi). In the laser conditions with low intensity of electric field (i.e., low laser power) or thermal energy (i.e., too fast scan speed), the transition to β-phase was negligible compared to optimum conditions (Figure S7, Supporting Information). Meanwhile, the LIPTα can be assumed to be a thermodynamics-dominant case with strong heat treatment to the PVDF microstructure. Excessive thermal energy allowed the microstructure of the PVDF to find the lowest energy state regardless

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of the accompanying electric field (Figure 1bii). Furthermore, the temperature where PVDF could be increased through laser was also an important issue since the excessively high temperature was expected to form α-phase, whereas moderate temperature for β-phase. However, it was hard to empirically measure the instant surface temperature of PVDF since the laser passed at ultrafast speed. Therefore, to compare the maximum temperature for LIPTα and LIPTβ, we analytically derived the maximum temperature of PVDF based on the first law of thermodynamics (Note S1, Supporting Information). Temperatures where phase transition occurred from γ- to β-phase were in the range of 101.21–150.41 °C. In the case of LIPTα, high temperatures over 216.93 °C which were higher than the melting point of PVDF were obtained. It would cause the randomization of the polarity of PVDF (i.e., β- to α-phase transition) (Figure S7, Supporting Information, color scale).

One of the major advancements of the LIPT over all previous studies is the ultrafast production of β-phase with high selectivity and precision. So far, most previous studies utilizing the laser have only been at the level of producing α-phase, thus the photothermal annealing effect has been only considered.[24–27] In contrast, ultrafast production of β-phase through the LIPTβ was possible by deeply considering the electric field of the laser along with the thermal effect. To elucidate our hypothesis, four comparative experiments were conducted which consists of (i) bulk thermal annealing (150 °C), (ii) laser scan without AuNPs, (iii) laser scan without AuNP accompanied by bulk annealing (150 °C), and (iv) laser scan with AuNPs (i.e., LIPTβ). The peak shift of Fourier-transform infrared spectroscopy (FT-IR) was investigated after each process (Figure S8, Supporting Information). Processes of (i) bulk thermal annealing and (ii) scanning laser without AuNPs showed no significant peak shift from α- to β-phase. It implies that the production of β-phase was impossible in the absence of either electric field or thermal annealing. When the (iii) laser was irradiated with 150 °C heat treatment, a slight peak shift was observed. It can be interpreted that enhanced chain mobility by bulk annealing gave more chances for the PVDF molecules to be aligned to an electric field from the laser. Finally, (iv) the LIPTβ allowed a clear peak shift from α- to β-phase. Such a transition can be ascribed to two major reasons. First, simultaneous exertion of electric field and thermal effect by the CW laser enabled the rearrangement of the PVDF molecules. Second, AuNPs played a crucial role to enhance them in an ultra-short reaction time.

The LIPT allowed monolithic integration of three main phases (i.e., α-, β-, and γ-phase) in the single PVDF layer which can design noble electromechanical responsibility. Monolithic phase integration of phases was confirmed with the microscope FT-IR mapping (Figure 1c). As prepared γ-phase PVDF was treated by single scanning of LIPTβ to patten β-phase and single LIPTα was overlapped on it for α-phase formation (Figure 1c, inset). The relative intensity of each phase (R_α for 614 cm^{-1}, R_β for 1279 cm^{-1}, and R_γ for 1234 cm^{-1}) was calculated point-by-point mapping in the measured area referring to previous studies.[30,31] The detailed procedure of calculating relative phase contents of PVDF was explained in Note S2, Supporting Information. The mapping results showed high β-phase contents in the area where LIPTβ was scanned, high α-phase contents for LIPTα processed area, and high γ-phase contents for the non-treated region. The process resolution of LIPT could be also identified as 20 μm for LIPTβ and 40 μm for LIPTα due to the difference in heat diffusion as can be seen through phase mapping results.

2.2. Computational Interpretation of Phase Transition Mechanism via MD Simulations

We carried out all-atom MD simulations to support the phase transition mechanism of the PVDF induced by laser irradiation (**Figure 2**a). A periodic cubic unit cell containing five randomly distributed α-phase PVDF molecules (100 repeat units for each chain) was prepared to study the bulk structure of materials (Figure 2ai). The initial thermodynamic equilibrium of the model and subsequent laser irradiation conditions are presented in Note S3, Supporting Information, Figures S9, and S10, Supporting Information. In this study, we characterized the difference in the laser scan speed as the relative difference in the heat and electric fields received by the area that has been exposed to the laser. In this regard, the ultrafast laser scan (i.e., LIPTβ) was modeled as an electrostatics-dominant field which consists of a fast oscillating electric field with mild heat pulse (±100 MV m^{-1} with 300–350 K heat pulse). Meanwhile, the thermodynamics-dominant field was modeled for slow laser scan (i.e., LIPTα) with strong heat pulse conditions (300–600 K heat pulse) without an electric field. By applying the two independent fields alternately, the change of dihedral angle distribution for each chain conformer (180° for trans conformers and 60, 300° for gauche conformers) was obtained. Similarly, the probability of trans conformer (R_t) was also computed to quantify the degree of phase change during the simulation (e.g., R_t = 0.5 for pristine α-phase, TGTG′). The MD simulation results indicated a definite increase in trans conformers (R_t ≈0.7) in the electrostatics-dominant environment (Figure 2aii,b), whereas the normal heat treatment (300–350 K heat pulse) did not change conformers (R_t ≈0.6) (Figure 2ai,b). It implies that the alternating electric field induced the structural rearrangement due to the strong dipole of hydrogen and fluorine atoms under increased chain mobility by the heat treatment. In opposite, under the thermodynamics-dominant field, the molecular structure was relaxed to a minimum-energy state. The restoration of gauche conformers was observed that potentially drives the reformation of α-phase (R_t ≈0.6) (Figure 2aiii,b).

The three Gaussian distributions centered on the average torsional angle of trans or gauche conformer were clearly found (Figure 2c). The PVDF backbones under electrostatics-dominant field (105 cycles including thermal cycling) showed intense distribution near 180°, indicating a formation of trans conformers. On the other hand, the conformer conversion to the gauche structure occurred in the thermodynamics-dominant environment, raising the probability density around 60° and 300°. For better comprehension, we compared the dihedral angle distribution over time when a single heat pulse was applied to the model under each field. Immediately after entering the electrostatics-dominant field, the probability density of the three peak regions (60°, 180°, and 300°) decreased due to fluctuations, and the unstable conformers were formed

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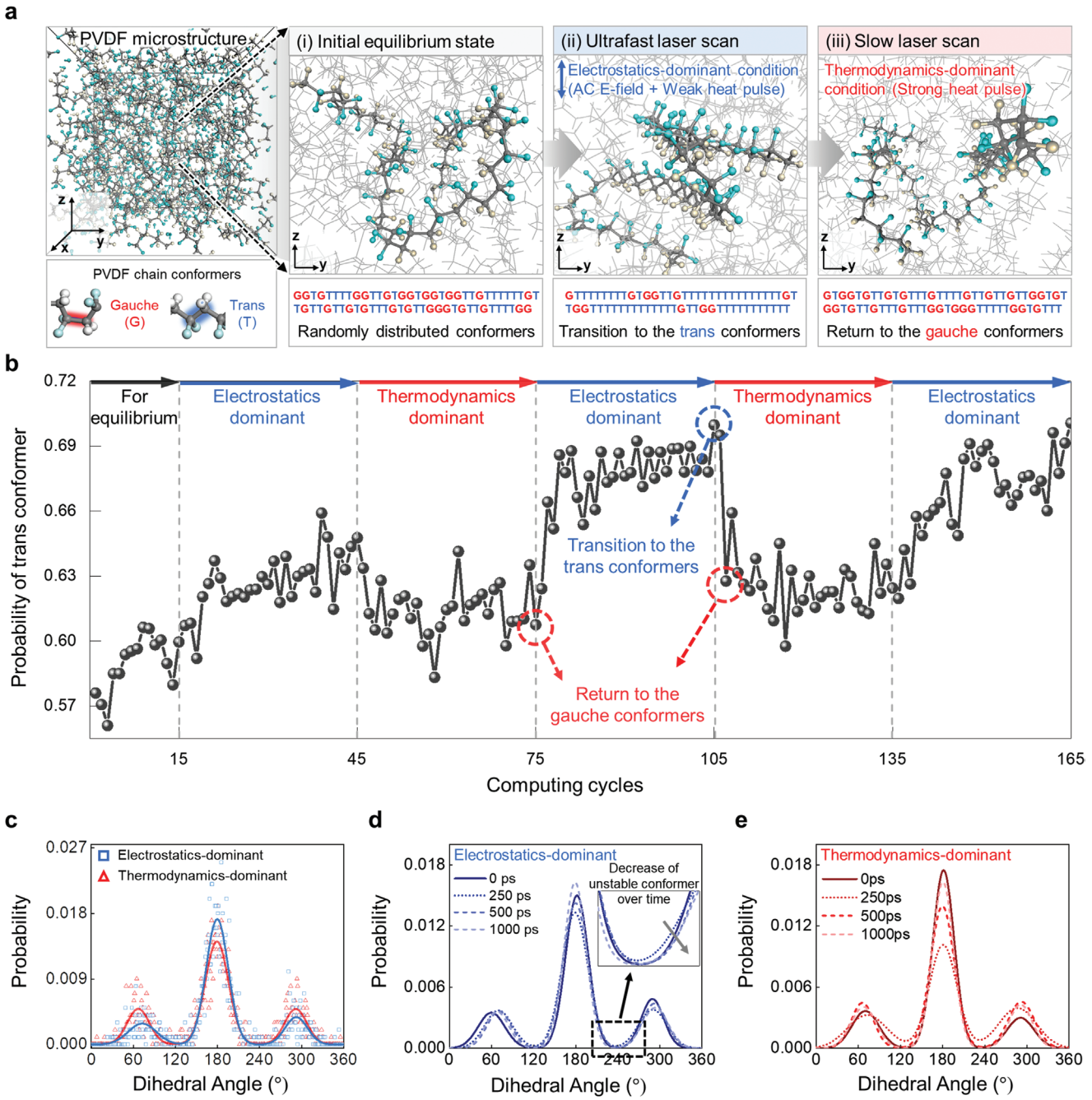


Figure 2. Computational interpretation of phase transition mechanism via MD simulations. a) Periodic cubic unit cell to study the bulk structure of PVDF under the simulation, PVDF microstructure at i) initial thermodynamic equilibrium state, ii) after LIPTβ, and iii) after LIPTα are depicted. Detail chain conformation is denoted by the serials of trans (T in the blue letter) and gauche (G in the red letter) conformers. b) The probability of trans conformers computed for quantification of phase change during the simulation, each modeled field was repeated for 30 cycles. c) The three Gaussian distributions centered on the average torsional angles of conformers. d,e) The change of dihedral angle distribution over time under a single heat pulse to the modeled field.

near 120° and 240°. However, the unstable conformers were mostly converted into trans conformer as cooling down proceeded under the AC electric field. Eventually, the probability density near 180° became higher than the initial state (Figure 2d). In contrast, the high thermal energy from the thermodynamics-dominant field allowed the molecular structure to overcome the energy barriers of the meta-stable state. A large number of trans conformers were converted into gauche or unstable conformers to form the lowest energy morphology. After cooling down to 300 K, the unstable conformers formed at high temperature were randomly re-distributed into the trans or gauche structure, having the three peak points of dihedral angle distribution increase simultaneously (Figure 2e).

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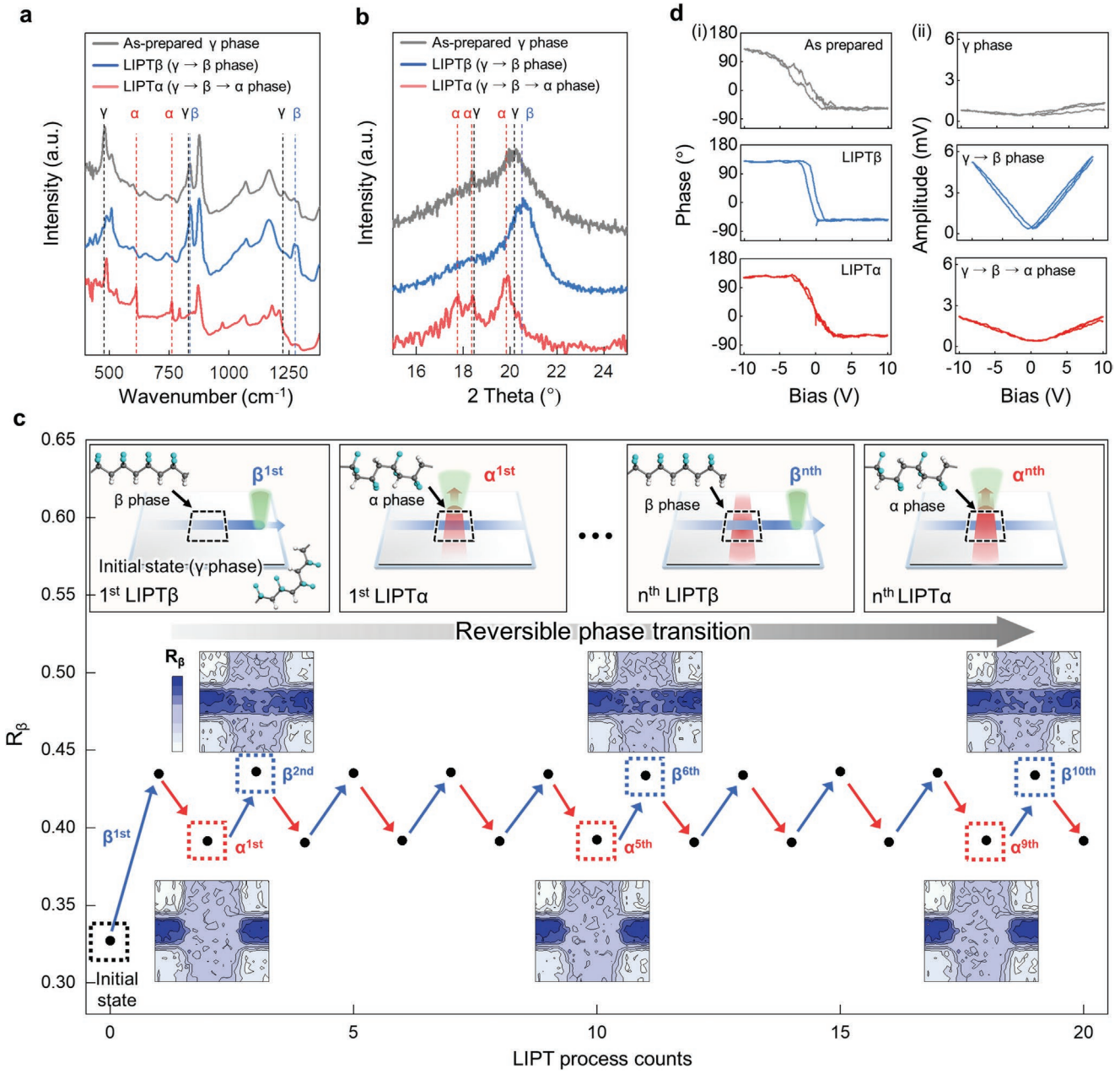


Figure 3. Material analysis on the phase transition of PVDF. a,b) Precise identification of polymorphic phase through FT-IR and XRD, clear peak shifts were observed as the LIPTβ and LIPTα proceeded. c) Quantitative analysis of reversible LIPT by calculating relative β-phase ratio in the area where alternative LIPT processes were treated. The flow of sample preparation is illustrated in inset schematics (top inset). Microscope FT-IR mapping also verified the selective and reversible phase transition (bottom inset). d) Piezoresponsive force microscope (PFM) analysis for a microscopic ferroelectric property measurement of each phase. The changes of i) PFM phase and ii) amplitude with PVDF phase transition were studied by sweeping DC bias from −10 V to +10 V.

2.3. Material Analysis on the Phase Separation of PVDF

Precise identification of the polymorphic phase was confirmed through FT-IR and X-ray diffraction (XRD)[6,7] (**Figure 3**a,b). Due to the similar chain conformation of three phases, it is difficult to strictly identify each phase using a single analysis technique (e.g., a similar peak assigned to β- and γ-phase in FT-IR, and α- and γ-phase in XRD). First, the prepared γ-phase was examined. The common peak of γ- and β-phase near 840 cm^{-1} was observed as expected in FT-IR. But exclusive peaks of γ-phase at 482, 1234 cm^{-1} and the absence of an exclusive β-phase peak at 1279 cm^{-1} confirmed it as a γ-phase dominant state. Further validation was conducted through XRD which can distinguish γ-phase at 20.04° from β-phase at 20.61°. The transformation of γ- into β-phase by the LIPTβ was confirmed through an enhanced peak at 840 cm^{-1} and a formation of the exclusive peak for β-phase at 1279 cm^{-1} in FT-IR. XRD ascertained the transition through the peak shift from 20.04 to 20.61°. The

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dramatic decrease of 840 cm^{-1} and formation of exclusive peaks of α-phase at 614 and 766 cm^{-1} in FT-IR convinced the formation of high-contents α-phase after treating the LIPTα. The peak shift from 20.61 to 19.9°, and the formation of an exclusive peak of α-phase at 17.66° also proved the phase transition by LIPTα. A similar peak of γ- and α-phase near 18° was shown that contributed by the gauche conformers[7] (γ-phase: 18.50°, TTTGTTTG′, and α-phase: 18.30°, TGTG′).

To verify the reliability and reversibility of the LIPT, the repeat test was performed by scanning the LIPTβ and LIPTα alternately ten times (Denoted by β^{nth} and α^{nth}) starting on the prepared γ-phase (Figure 3c, inset schematics). Since AuNPs were ablated after the LIPTα processes, AuNP inks were re-coated before the following LIPTβ processes. The relative intensity of the β-phase (i.e., R_β) from FT-IR was used to compare the phase composition. The R_β value was calculated on the overlapped area where the LIPTβ and LIPTα have been processed alternately (Figure 3c, inset schematics, dotted box). A sharp increase of the R_β was obtained by first LIPTβ (β^{1st}), then highly reversible R_β value was repeated by each cycle. The microscopic FT-IR mapping also verified the repeatability and precision (Figure 3c, inset mapping images).

Besides the identification of the phase change, the microscopic ferroelectric switching and electromechanical behavior of the PVDF were measured through a piezoresponse force microscope (PFM)[32–34] (Figure 3d). Each phase of the PVDF was prepared on the conducting substrate, and an atomic force microscope (AFM) cantilever with a conductive tip was scanned with vertical mode for measurement. The DC voltage was reversed from −10 V to +10 V to observe the ferroelectric phase switches of 180° (Figure 3di). Despite a little hysteresis loop found from γ-phase, the coercive field was not definite due to unstable dipole arrangement. An ideal domain switching was observed for the LIPTβ sample with −0.141 V and −1.08 V of the coercive field. Such a small and clear coercive field was contributed by high β-phase contents where the electric dipole of polar chain conformation was easy to be rotated in response to a sweeping DC bias.[32] The hysteresis loop disappeared after treating LIPTα indicating its nonpolar α-phase formation. Meanwhile, the amplitude signal versus DC bias revealed the local strain of the PVDF induced by the PFM tip (Figure 3dii). Unlike the prepared γ-phase and LIPTα, the LIPTβ sample showed not only a clear butterfly loop but also high amplitude induced by the active expansion and contraction of the dipoles in the β-phase PVDF. The abrupt change of the strain was observed at −0.85 V and 0.35 V which implicates the fast reactivity of polar structure in β-phase. We also calculated the piezoelectric constant d33 through PFM results where LIPTα showed 29.71 and 41.48 pm V^{-1} for LIPTβ. Interestingly, LIPTα has notable amplitude in PFM data. For deduction of this result, we characterized the quantitative crystallinity of each phase (i.e., as prepared γ-phase, LIPTβ, and LIPTα) by deconvolution of XRD data (Figure S11, Supporting Information). The diffraction curves were divided into two regions, crystalline (solid lines) and amorphous (short dash line). The degree of crystallinity (Xc) was calculated from the ratio of crystalline to total area (crystalline + amorphous) under the diffraction curves. Through photo-thermal annealing of LIPT processes, the crystallinity of PVDF was increased after laser treatment. The crystallinity of the as-prepared γ, LIPTβ, and LIPTα were 46.21%, 56.42%, and 55.28%, respectively. Therefore, the enhanced crystallinity with successive LIPT processes could increase the PFM amplitude.

2.4. Steerable Soft Robot Enabled by Monolithic Phase Integration of PVDF

To verify the potential of the LIPT as a means of electromechanical device production, a uni-morph soft robot with a practical function was demonstrated (**Figure 4**a). First, we prepared cantilever-type flat uni-morphs to compare the difference in resonant frequencies and vibrating amplitudes between LIPTα and LIPTβ. PVDF layer was sandwiched by the conducting polymer, (poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate), (PEDOT:PSS)) electrodes for facile and conformal coating. Polyimide (PI) was used for the non-deforming substrate layer (Figure 4ai, and Figure S12, Supporting Information). The distinct difference in the resonant frequency of each phase was confirmed by measuring the amplitudes of single-phase, uni-morph cantilever actuators through digital snapshots (Figure S13, Supporting Information). The monolithic phase integration (α- and β-phase) was formed by treating LIPTα and LIPTβ half and half. The curved structure was designed for free movement of the robot, and we added the function of direction shift by integrating heterogeneous phases in a single PVDF through the digital laser patterning process, which is not reported in piezoelectric actuators before. The soft robot was able to vibrate through the alternating contraction and extension of the PVDF by the AC electric field while not deforming the PI layer (Figure 4aii). The various movement was observed depending on driving frequency. Owing to the mixed electromechanical response of the PVDF layer integrated with dual phases, it had unique resonant frequencies for each phase making direction control of the soft robot (e.g., left turn, moving forward, and right turn) (Figure 4aiii).

The trajectory of the soft robot was captured with varying driving frequencies. Resonant frequencies were identified as 40 Hz for β-resonant which made a left turn of the soft robot, 50 Hz for α-resonant making a right turn, and 30 Hz of central frequency for a straight movement owing to similar vibration amplitude of both phases (Figure 4b,c). The reliability of the soft robot was manifested by measuring the curvature radius of the trajectories, where β- and α-resonant resonant frequencies showed 10 repeated outputs (Figure 4c,d, and Figure S14, Supporting Information). The speed of the soft robot was determined by driving frequency. The β-resonant frequency allowed faster movement than the α-resonant one owing to the higher electromechanical response (Figure 4dii). Finally, to demonstrate a practical application, the soft robot followed the planned target path by controlling the direction on-demand (Figure 4e and Video S1, Supporting Information). The soft actuator started moving to the left direction with 40 Hz of the driving frequency. A dramatic change of direction occurred as we changed the frequency to 50 Hz which was α-resonant. The soft robot recovered its direction when the frequency was changed to 40 Hz again.

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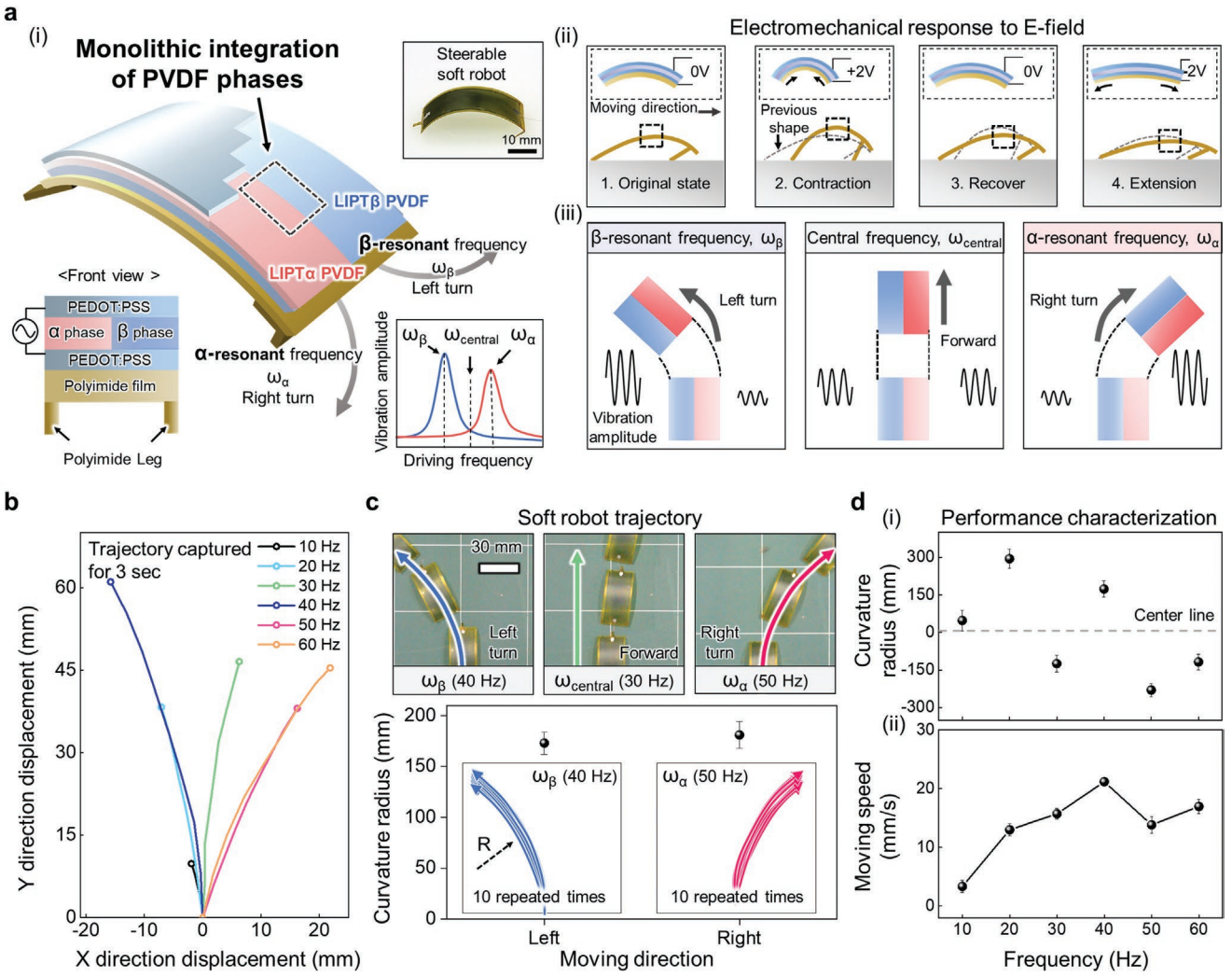


Figure 4. Steerable soft robot enabled by the monolithic phase integration of PVDF. a)Schematic illustration of a uni-morph soft robot with monolithically integrated PVDF phase (α- and β-phase) layer. i) The soft robot consists of two electrode layers and a single substrate layer (left bottom schematics). There exist separated resonant frequencies for each phase (right bottom graph). ii) The moving mechanism of the soft robot by the alternating electric field to the PVDF layer. iii) Direction controllability of the soft robot by resonant frequencies. b,c,d) Characterizations of the soft robot movement depending on the driving frequencies (trajectory curvature radius and speed). e) Demonstration of the practical soft actuator applications following planned target path by controlling its moving direction. Values in c,d represents the mean and the error bar represents the standard deviation ($n = 10$).

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3. Conclusion

We present an on-demand, ultrafast, and reliable phase-control technique utilizing the phase transition mechanism induced by a CW laser with the aid of plasmonic nanoparticles. Contrary to the conventional multi-step processes, the LIPT successfully transformed three polymorphic phases without giving film deformation. As we verified through a steerable soft robot, the selectivity of LIPT can optimize and customize the functions of electromechanical devices by programming each phase in a single PVDF which is not allowed in conventional approaches. Although laser-induced photo-thermal heating has limitations in a temperature gradient in the depth direction, it is enough to treat the thin film of the target polymer and is likely to be solved by mixing nanomaterials with polymer varnish. The LIPT promises an efficient strategy for relevant research areas where the molecular structure with polarization is of great interest. We expect the concept of utilizing the plasmonic nanomaterials to amplify the characteristics of laser for molecular structure change is expanded to emerging functional electronic materials. We also believe that future laser process research will not only observe the apparent behavior of materials but more suggest a systematic design method based on underlying mechanics, which is essential for broadening the usability of the associated system.

4. Experimental Section

Material Preparation: The PVDF powder (average M_w ≈534 000) was dissolved in N, N-dimethylformamide (DMF) to prepare a 15 wt% of varnish. The varnish was spin-coated at 1,000 rpm for 30 s on the glass substrate to prepare the γ-phase PVDF thin film. The film thickness of prepared γ-phase PVDF was adjusted to 10 µm. 0.1 wt% of AuNP solution (PVP-stabilized and dispersed in ethanol; Ditto Technology) was spin-coated on the prepared PVDF for laser absorption layer. PEDOT:PSS aqueous solution (1.0–1.3 wt% solid content, CleviosTM PH1000; Heraeus Electronic Materials) was coated on both sides of PVDF for electrode layers to fabricate uni-morph actuators.

Laser-induced Phase Transition of PVDF (LIPT): The laser scanning system consisted of a CW 532 nm laser (Sprout-G-5 W; Lighthouse Photonics), f-theta telecentric lens (f = 103 mm), and Galvano-mirror (hurrySCAN II; Scanlab), which was used for laser processing to induce selective phase transition (Figure S15, Supporting Information). In this laser system, it has been configured a low numerical aperture (N.A. ≈ 0.017) system to obtain a long DOF that was sufficient to cover the 10 µm thickness PVDF film. The focused beam size ($k^*\lambda$/N.A., $k = 2/\pi$ = constant, λ = 532 nm) was about 20 µm and DOF ($k^*\lambda/(N.A.)^2$) was hundreds of micrometers which is variable with laser beam quality (M^2). LIPT with high DOF could evenly treat the PVDF to the depth direction. For phase mapping through microscope FT-IR, a single laser scan of LIPTβ and LIPTα was performed. For all other sample preparation for material characterizations and actuator fabrications, large area scanning of LIPT was conducted for phase transformation by setting the hatch size to 20 µm for LIPTβ and 40 µm for LIPTα which were the spatial resolutions for each process.

MD Simulation Study: The 3D all atomic structures were modeled in commercial software Materials Studio (Dassault Systèmes, France), and simulation of the laser environment including heating and the electric field was carried out with open source code LAMMPS (Sandia Lab., USA).

Microscopy: The surface morphology was characterized by a scanning electron microscope (FE-SEM, Auriga; Carl Zeiss) and AFM (NX 10; Park Systems). Piezoelectric response of each phase was characterized by the additional accessory to AFM (NX 10; Park Systems).

Material Characterization: The optical property of the PVDF film was analyzed with a UV-vis spectrophotometer (V-770, Jasco). The identification of each phase was analyzed with XRD (New D8 Advance, Bruker), FT-IR (Nicolet iS50, Thermo Fisher Scientific), and microscope FT-IR (Nicolet iN10, Thermo Fisher Scientific).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

D.W. and H.C. contributed equally to this work. D.W., H.C., S.W.K, and S.H.K. designed the project and experiments. D.W. and H.C. designed and carried out the experiments. H.K. and J.C. designed and performed molecular dynamics simulations. S.H. provided ideas for MD simulation. G.L. characterized the optical property of materials. J.K., J.K., and S.H. provided ideas for development. J.C., S.W.K. and S.H.K. supervised the research program. All authors discussed the results and wrote the manuscript.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

electromechanical applications, laser-induced phase transition, monolithic phase integration, piezoelectric polymers, plasmonic nanoparticles, soft robots, ultrafast and selective phase change

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